

Stat 101

Section 1-3 Worksheet

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1.26

The population in a public university is 7,000 students and you want to sample 200 students

How would you sample these students systematically ?

Given information:

The population size $N = 7000$

The sample size is $n = 200$

In systematic sampling method:

Step1: Divide the population to 200 groups with $k = \frac{7000}{200} = 35$ ordered students in each group.

Step 2: From the first group we select one student randomly (Named a start point)

Step3: Select every 35th student thereafter until 200 students are sampled

Note:

The range of the first one we selected is from 1 to 35

1.27

Describe the difference between a statistic and a parameter.

A parameter is a descriptive numerical measurement such as an average, proportion that is calculated from the entire population

A statistic is is a descriptive numerical measurement such as an average, proportion that is calculated from a subset of the population (Sample)

1.28

Why is convenient sampling is considered to be non-statistical sampling method?

Because: In convenience sampling

- 1. Samples are chosen by easy or convenient way for the researcher (Reason not randomly)**
- 2. Not every item in the population has the same chance to be selected.**

Thus, convenience sampling is not a statistical sampling method.

1.29 (Similar to 1.26)

Describe how Kris plans to use the systematic sampling method to sample 100 securities listed on the New York Stock Exchange.

Assume that the wall street Journal has 2,531 securities listed in the New York Exchange Composite Transactions” in an alphabetical order

Given information:

The population size $N = 2531$

The sample size is $n = 100$

In systematic sampling method

Step1: Divide the population to 100 groups with

$k = \frac{2531}{100} = 25.31 = 25$ ordered securities in each group

Step 2: From the first group we select one security randomly

Step3: Select every 25th student thereafter until 200 students are sampled

Note:

The range of the first one we selected is from 1 to 25

1.34

The Human resources manager of a university e-mailed a survey questionnaire to his 150 employees in the company

The sample included 50 employees randomly selected from each of the category of administration, teaching staff, and non-teaching staff

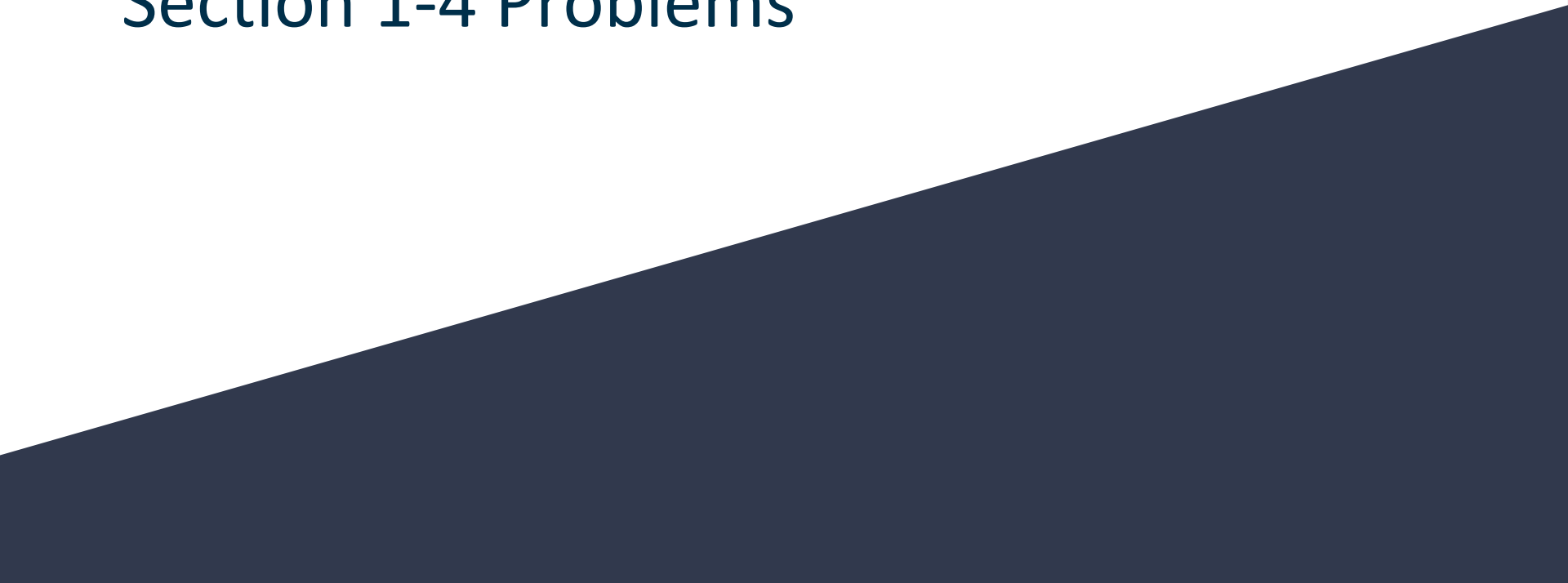
What sampling method was used by the manager? Explain

The manager used the stratified random sampling method

Because:

- 1- The entire university was divided into three subgroups, administration, teaching staff and non-teaching staff (**Depend on common characteristic**)
- 2- The items of the sample are selected from each group

Section 1-4 Problems

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1.43

What is the difference between qualitative and quantitative data

Qualitative data, we can categorize the data to categories represented by names (Characteristics for example color) or by numbers (codes for example Students ID).

Quantitative data is numerical. It has two types

1. Continuous (numbers that could have decimals and to find them you need to measure them by tool or formula)
2. discrete (exact number (integer number) and to find them you need to count)

1.42

For each of the following indicate whether the data are cross sectional or time-series

A- quarterly unemployment rate

Time – series (key word quarterly means in each 4 month)

B- unemployment rates by state

Cross sectional

C- monthly sales

Time – series (key word monthly means in each month)

D- employment satisfaction data for a company

Cross sectional

1.49

The following information on 15 different state investment trusts has been collected. Identify whether the data are cross sectional or time-series

A- Income distribution by region in 2016

Cross sectional

B- Per share funds operation for the years 2012-2016

Time-Series

C- Number of residential apartment buildings owned on December 31, 2015

Cross sectional

D- Overall percentage of leased space for the 137 properties in services as of December 31, 2015

Cross sectional

E- Annual dividend paid to shareholders in 2010-2015

Time-Series

1.45

What is the difference between nominal and ordinal data

Nominal data involves placing observation in separate categories according to some measurable characteristic. (No meaning for the rank)

Ordinal data also involves placing observations into separate categories, but the categories can be rank-ordered

1.44

For each of the following variable indicate the level of data measurement

A- Product rating (1- excellent, 2- good, 3=fair, 4=poor, 5= very poor)

Ordinal-categories with defined order

B- Home ownership (own, rent, other)

Nominal- categories with no defined order

C- College grade point average

Ratio

D- marital status (single, married, divorced, other)

Nominal- categories with no defined order

1.47

Company's HR department wants to pick a sample of employees from total of 500 employees in the company and ask how they feel about the company policies.

The employee's list in alphabetical order is used for the study where every 10th employee will be selected to the sample

A- What is the population of interest

The population of interest is the entire company of 500 employees

B- What is the sample

The sample is the 50 selected employees obtained by choosing every 10th employee from the alphabetically ordered list of employees

$$\frac{N}{n} = k \rightarrow \frac{500}{n} = 10 \rightarrow n = \frac{500}{10} = 50$$

C- What is the sampling method being used

The systematic sampling method is being used